

1. Embedding design

To convert structured funding-related data into semantically rich vector embeddings using a pre-trained SentenceTransformer model. These embeddings will be used for similarity search, recommendation, or clustering tasks.

2. Input

A `.jsonl` (funding_finetune_data.jsonl)

The JSONL file is essentially a list of user-question → ideal-answer examples covering your funding data. Each line in that file has:

A “prompt” (the user’s hypothetical query), such as “Which grants offer marine biodiversity funding?”

A “completion” (the correct or desired reply), referencing columns like Funding Amount, Application Deadline, Eligibility Criteria, etc.

So, intuitively, the JSONL file is a set of labeled examples (prompts and completions) that show the LLM how to respond to real-world questions about your grants.

Example:

```
prompt_1 = f"We are looking for funding focused on {funding_focus}. Does {funder} fit our needs?"
```

```
completion_1 = f"{funder} offers {funding_amount}. The application deadline is {application_deadline}. ..."
```

3. Output

A new `.jsonl` (data_with_embeddings.jsonl) file where each item includes an additional "embedding" field with its corresponding vector representation.

Example:

```
{
  "Funder": "ABC Foundation",
  "Type": "Private Fund",
  "Funding Amount": "$10,000 - $50,000",
  "Application Deadline": "2024-12-31",
  "Funding Focus": "Climate and sustainability",
  "Eligibility Criteria": "NGO"
  "embedding": [0.123, -0.456, ...]
}
```

4. Embedding Pipeline Architecture

1. Data Loading

The input file is read line-by-line with UTF-8 encoding. Each line is parsed as a JSON object.

Invalid lines are skipped with warnings.

2. Text Construction

For each record, the relevant fields are extracted.
Fields are concatenated using " | " as a separator.

3. Model Inference

Sentence embeddings are generated using all-MiniLM-L6-v2 from sentence transformers.
Embeddings are computed in batches with a progress bar.

4. Data Augmentation

Each record is augmented with its corresponding embedding (as a list of floats under the key "embedding").

5. Output Writing

Enriched records are written to data_with_embeddings.jsonl in JSONL format.

5. Dependencies

sentence-transformers is a Python library built on top of Hugging Face's transformers.

It's designed specifically for:

Sentence Embeddings: Turning full sentences or paragraphs into meaningful vectors.

Tasks like:

Semantic search

Clustering

Duplicate detection

Sentence similarity